

Activity 6 Maxine's car

Aim: Build spreadsheets as a financial model.

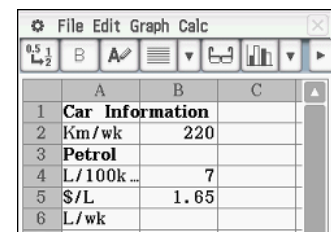
Maxine wants to buy a car and wants to make sure she can meet the costs of running and maintaining the car. She can afford up to \$50/week.

Her research is summarised in the table below.

Item	Assumptions
Usage	220 km per week
Fuel (petrol)	Fuel is currently \$1.65 / litre. The car uses 7 litres every 100 kilometres.
Tyres	A new set costs around \$390 and lasts 50 000 km
Service	\$150 every 5000 km
Repairs	\$500 per year
Registration	\$650 per year
Insurance	For third party property it is \$235 per year

1. How much will Maxine pay for petrol each week?
 - a) How many litres are required for 220 km?
 - b) How much does that quantity of petrol cost each week?

2. Build a spreadsheet to do this calculation.



	A	B	C
1	Car Information		
2	Km/wk	220	
3	Petrol		
4	L/100k...	7	
5	\$/L	1.65	
6	L/wk		

- a) What is the formula in cell B6?
 - b) Compare your answers to Q1 a) and Q2 a)
3. If Maxine chooses a more fuel efficient car that only uses 4.6 L per 100 km, what will petrol cost per week?
(Hint: Tap in cell B4 and change the number. Change it back before answering Q4.)

4. Modify the spreadsheet as shown.

- (Use the information shown in the table on the previous page.)
 - Cells B13 to B19 each require a formula.
- a) Use the spreadsheet to determine Maxine's total running costs per week.
- b) Hence determine whether or not she can afford the car.

	A	B	C
5	\$/L	1.65	
6	L/wk	15.4	
7	Tyres	390	50000
8	Service	150	5000
9	Repairs	500	
10	Registr...	650	
11	Insura ...	235	
12			
13	Petrol	25.41	
14	Tyres	1.716	
15	Service	6.6	
16	Repairs	9.61538	
17	Registra...	12.5	
18	Insurance	4.51923	
19	Total \$	60.3606	
20			

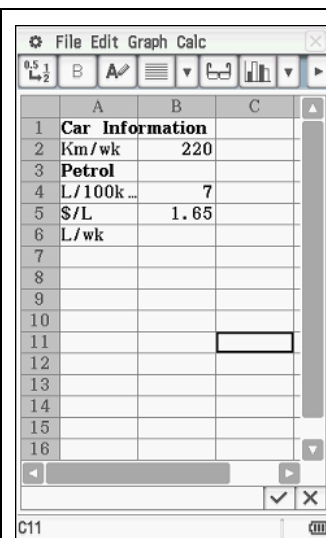
5. Research a car of your own choosing, adjust the values in your spreadsheet and determine the running cost per week.

Learning notes

Detailed instructions for complete spreadsheet

Build the spreadsheet shown.

- Tap , tap the Spreadsheet button
- Tap in cell A1
- Enter a heading Car information
- Tap  to make bold
- Tap in cell A2 and enter Km/wk
- Tap in cell A3 and enter Petrol
- Tap in cell A4 and enter L/100km
- Tap in cell A5 and enter \$/L
- Tap in cell A6 and enter L/wk
- Tap in cell B2 and enter 220
- Tap in cell B4 and enter 7
- Tap in cell B5 and enter 1.65

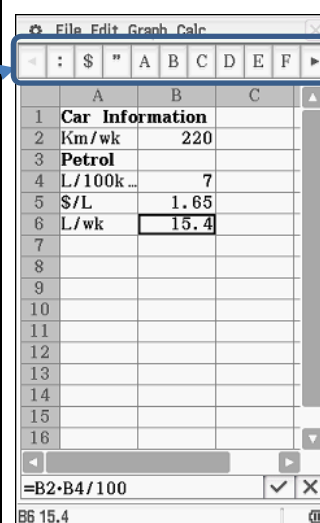


	A	B	C
1	Car Information		
2	Km/wk	220	
3	Petrol		
4	L/100k ...	7	
5	\$/L	1.65	
6	L/wk		
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

Enter formulas

To enter a formula you must start with an =

- Tap in cell B6
- Press  (the menu bar changes)
- tap cell B2, press , enter 100, press , tap in cell B4 and press  (divide the 220 by 100 to calculate how many hundreds of kilometres and then multiply by the number of litres used per 100 km.)
- Tap in cell D6
- Press , tap cell B6, press , tap in cell B5 and press .



	A	B	C
1	Car Information		
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3	Petrol		
4	L/100k ...	7	
5	\$/L	1.65	
6	L/wk	15.4	
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

To convert cost per year to cost per week divide by 52.